

Case Study Of AREN'S FANCY WORLD

Mid-Term Report for Business Data Management Capstone Project

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EXECUTIVE SUMMARY:-

Aren's Fancy World is a gift shop, stationery shop, and fancy dress competition costume vendor. In a well-established township, the business enjoys access to a customer base of students, parents, and event organizers. Since the business is well established, its operational issues are prominent with huge effects on profitability and efficiency.

The biggest challenges are in controlling stock, holding on to customer loyalty, and handling returns by customers as well as special orders. The greatest issue is customers returning merchandise once they purchase the items, which affects turnover in stock. In addition, customer demands for special orders whenever products are unavailable frequently force the owner to buy the products in large quantities, thereby causing the issues of selling the excess stocks. In some cases, consumers make orders for merchandise but fail to purchase it, causing financial woes and inventory overflow.

In order to respond to these questions, I undertook a critical analysis through questioning previous purchase records, selling records, and Excel sheets as well as with a short interview of the owner. The aim is to design a formal inventory control system that optimizes stock levels, minimizes wastage, and maximizes order fulfillments from the customers.

With the help of data-based insights, this project aims to suggest better inventory tracking practices, a return policy framework, and an ordering policy that has been optimized to reduce monetary losses and enhance business efficiency. By implementing these solutions, Aren's Fancy World can maximize profitability, offer better stock control, and retain customers better.

PROOF OF ORIGINALITY:-

- **Business Name:** Aren's Fancy World
 - **Owner's Name:** Mr. Amit Aren
 - **Business Address:** Orbit Plaza SHOP NUMBER UG 25 GROUND FLOOR ORBIT PLAZA, near AXIS BANK ATM, Crossings Republik, Ghaziabad, Uttar Pradesh 201016
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1. **Interaction Video with Owner:**
https://drive.google.com/file/d/1Wm87ODfr2EZQ_WUKd_dI19HqwmM3HUR8/view?usp=sharing
 2. **Letter from Owner (NOC):**
https://drive.google.com/file/d/11Yq6m6-z78QByWT4M8WzIIAZ_U_KfkUw/view?usp=sharing

AREN'S FANCY WORLD

No Objection Certificate

Date: 29-03-2025

This is to certify that Mr. Bhavya Singh has taken the data from our organisation and this letter confirms that we have completely no objection to his using our data for his BDM Capstone project.

This **NOC** has been issued as per the request of Bhavya Singh and can be used by him for the specific purpose mentioned above.

During his time with us, he has been a sincere individual and we wish him good luck for his project.

Sincerely,

Mr. Amit Aren



3. Data (Google Sheets):

https://docs.google.com/spreadsheets/d/11mlUeu9Sstd1Ej7OOtSApFVje35hIbAvFuH_pt_Gn64/edit?usp=sharing

4. Images of Shop:



https://drive.google.com/file/d/1F_F-Cq_54rWlaS7BfwcNT3PD7kxQKv7k/view?usp=sharing



<https://drive.google.com/file/d/1woDNytzX8y3MWYajT8TZNW-d49zRMq9L/view?usp=sharing>



https://drive.google.com/file/d/1_4l2xTr44WxdqcxjsFRs71WmfBPJENJY/view?usp=sharing

METADATA:-

- Data Format: Excel sheets
- Data Range: 1st November 2024 30th November 2024
- Data Collection Method: The data was recorded using Google Sheets, verbal notes. It was gathered on monthly based on sales and purchase and inventory, data was taken on the first and the last date of the month(1st November and 30th November respectively).
- https://docs.google.com/spreadsheets/d/11mlUeu9Sstd1Ej7OOtSApFVje35hIbAvFuH_pt_Gn64/edit?usp=sharing

The owner willingly gave me the access to the sales, purchase and inventory data.

1. **Sales data:** The owner shared an excel sheet comprising of the sales report containing details of specific item names, quantities sold, and the total revenue generated per item only for the stationary items sold by the shop.

Item	Units	Sales Price (per unit)	Overall Revenue
DOMS FSC 12 Shades Round Shaped Color Pencils	7	58	406
PIERRE CARDIN Estella Ball Pen	19	95	1805
Reynolds FASTLINE Ball Pen	168	7	1176
Reynolds Lumino Gel Gel Pen	189	10	1890
APSARA Disney-Marvel(Avengers, Disney Princess, Mickey Mouse) Rotating Dispenser of Pencil	2	465	930
FLAIR Woody 0.7 mm Stand Attractive Woody Design Smooth Ink Flow System Ball Pen	200	10	2000
CASIO MJ-12VCB-WE Desktop Basic Calculator	10	535	5350
CASIO MC-12M Basic Calculator	7	595	4165
CASIO MJ-120D Desktop Basic Calculator	8	540	4320
HAUSER Aerox 0.6 mm Ball Pen Shiny Finish & Low-Viscosity Ink Refillable Ball Pen	312	10	3120
Classmate Octane Gel Pen	500	10	5000
HAUSER XO 0.6 mm Sleek Body & Minimalistic Design Smooth Writing Ball Pen	197	15	2955
Reynolds HD Blue Permanent Marker	38	40	1520
Reynolds HD Blue Whiteboard Marker	83	35	2905

Item	Units	Sales Price (per unit)	Overall Revenue
Cello Butterflow Simply Ball Pen	300	10	3000
Classmate 04010030AA Geometry Box	31	110	3410
PARKER Classic Stainless Steel Gold Ball Pen	21	150	3150
LUXOR Executive Regular Notebook Single Rule 300 Pages	55	60	3300
LUXOR Premium Exercise Regular Notebook Ruled 300 Pages	60	70	4200
Reynolds Aeroslim Ball Pen	165	15	2475
Reynolds VISTA RT BP Blue Pen	461	20	9220
Reynolds DynaGrip Roller Ball Pen	134	25	3350
Reynolds Trimax Pen - BLUE, BLACK, RED & GREEN Gel Pen	171	55	9405
Reynolds Brite BP Pen Ball Pen	251	7	1757
Reynolds JIFFY Blue Gel Pen	197	10	1970
Reynolds Brite Blue Pen Packet Ball Pen	100	60	6000
Reynolds D Fine Ball Pen	188	30	5640

Item	Units	Sales Price (per unit)	Overall Revenue
Reynolds Feltek Roller Ball Pen	118	35	4130
Reynolds Trimax Pens Pouch - Blue, Black, Red & Green Gel Pen	21	150	3150
Reynolds SmothMate Ball Pen	86	35	3010
Reynolds 045 Pen Jar Ball Pen	6	80	480
Reynolds JIFFY Gel Pen	189	10	1890
FLAIR Silkina Jar of Ball Pen	4	150	600
Cello Butterflow Simply Ball Pen	350	10	3500
Classmate Pulse 6 Subject (240 X 180 mm) Spiral B5 Notebook Single Line 300 Pages	68	80	5440
Classmate Pulse 1 Subject (240 X 180 mm) Spiral B5 Notebook Single Line 180 Pages	70	55	3850
NAVNEET Youva Soft Bound Brown Note Book 18x24 cm Single Line 92 Pages Regular Note Book	32	35	1120
Classmate Pulse 1 Subject (267 X 203 mm) Spiral Regular Notebook Single Line 160 Pages	31	50	1550
Classmate Longbook A4 Notebook Unruled 140 Pages	57	40	2280
Classmate Longbook A4 Notebook Single Line 240 Pages	60	60	3600
CASIO FX-991MS Scientific Scientific Calculator	16	1200	19200

2. **Purchase data:** The owner shared an excel sheet comprising of the purchase report containing details of specific item names, units purchased, and the total cost per item only for the stationary items sold by the shop.

Item	#	Units	#	Price (per unit)	Overall Cost
DOMS FSC 12 Shades Round Shaped Color Pencils		10		60	600
PIERRE CARDIN Estella Ball Pen		20		80	1600
Reynolds FASTLINE Ball Pen		250		3	750
Reynolds Lumino Gel Gel Pen		200		7	1400
APSARA Disney-Marvel(Avengers, Disney Princess, Mickey Mouse) Rotating Dispenser of Pencil		4		480	1920
FLAIR Woody 0.7 mm Stand Attractive Woody Design Smooth Ink Flow System Ball Pen		200		5	1000
CASIO MJ-12VCB-WE Desktop Basic Calculator		10		555	5550
CASIO MC-12M Basic Calculator		10		595	5950
CASIO MJ-120D Desktop Basic Calculator		10		555	5550
HAUSER Aerox 0.6 mm Ball Pen Shiny Finish & Low-Viscosity Ink Refillable Ball Pen		500		5	2500
Classmate Octane Gel Pen		550		5	2750
HAUSER XO 0.6 mm Sleek Body & Minimalistic Design Smooth Writing Ball Pen		300		5	1500
Reynolds HD Blue Permanent Marker		50		25	1250
Reynolds HD Blue Whiteboard Marker		100		20	2000
Item	#	Units	#	Price (per unit)	Overall Cost
Cello Butterflow Simply Ball Pen		300		5	1500
Classmate 04010030AA Geometry Box		50		60	3000
PARKER Classic Stainless Steel Gold Ball Pen		25		110	2750
LUXOR Executive Regular Notebook Single Rule 300 Pages		60		40	2400
LUXOR Premium Exercise Regular Notebook Ruled 300 Pages		60		50	3000
Reynolds Aeroslim Ball Pen		200		7	1400
Reynolds VISTA RT BP Blue Pen		500		15	7500
Reynolds DynaGrip Roller Ball Pen		200		15	3000
Reynolds Trimax Pen - BLUE, BLACK, RED & GREEN Gel Pen		200		40	8000
Reynolds Brite BP Pen Ball Pen		300		2	600
Reynolds JIFFY Blue Gel Pen		200		5	1000
Reynolds Brite Blue Pen Packet Ball Pen		100		29	2900
Reynolds D Fine Ball Pen		200		20	4000
Reynolds Feltek Roller Ball Pen		150		25	3750
Item	#	Units	#	Price (per unit)	Overall Cost
Reynolds Feltek Roller Ball Pen		150		25	3750
Reynolds Trimax Pens Pouch - Blue, Black, Red & Green Gel Pen		50		109	5450
Reynolds SmothMate Ball Pen		100		28	2800
Reynolds 045 Pen Jar Ball Pen		10		37	370
Reynolds JIFFY Gel Pen		200		3	600
FLAIR Silkina Jar of Ball Pen		10		75	750
Cello Butterflow Simply Ball Pen		350		4	1400
Classmate Pulse 6 Subject (240 X 180 mm) Spiral B5 Notebook Single Line 300 Pages		80		38	3040
Classmate Pulse 1 Subject (240 X 180 mm) Spiral B5 Notebook Single Line 180 Pages		70		30	2100
NAVNEET Youva Soft Bound Brown Note Book 18x24 cm Single Line 92 Pages Regular Note Book		60		25	1500
Classmate Pulse 1 Subject (267 X 203 mm) Spiral Regular Notebook Single Line 160 Pages		70		20	1400
Classmate Longbook A4 Notebook Unruled 140 Pages		70		25	1750
Classmate Longbook A4 Notebook Single Line 240 Pages		60		30	1800
CASIO FX-991MS Scientific Scientific Calculator		20		1215	24300

3. **Inventory data:** The owner shared an excel sheet comprising of the items in the inventory along with the number of units of the stationary items left after the month.

Item	#	Units
DOMS FSC 12 Shades Round Shaped Color Pencils		3
PIERRE CARDIN Estella Ball Pen		1
Reynolds FASTLINE Ball Pen		82
Reynolds Lumino Gel Gel Pen		11
APSARA Disney-Marvel(Avengers, Disney Princess, Mickey Mouse) Rotating Dispenser of Pencil		2
FLAIR Woody 0.7 mm Stand Attractive Woody Design Smooth Ink Flow System Ball Pen		0
CASIO MJ-12VCB-WE Desktop Basic Calculator		0
CASIO MC-12M Basic Calculator		3
CASIO MJ-120D Desktop Basic Calculator		2
HAUSER Aerox 0.6 mm Ball Pen Shiny Finish & Low-Viscosity Ink Refillable Ball Pen		188
Classmate Octane Gel Pen		50
HAUSER XO 0.6 mm Sleek Body & Minimalistic Design Smooth Writing Ball Pen		103
Reynolds HD Blue Permanent Marker		12
Reynolds HD Blue Whiteboard Marker		17
Item	#	Units
Cello Butterflow Simply Ball Pen		0
Classmate 04010030AA Geometry Box		19
PARKER Classic Stainless Steel Gold Ball Pen		4
LUXOR Executive Regular Notebook Single Rule 300 Pages		5
LUXOR Premium Exercise Regular Notebook Ruled 300 Pages		0
Reynolds Aeroslim Ball Pen		35
Reynolds VISTA RT BP Blue Pen		39
Reynolds DynaGrip Roller Ball Pen		66
Reynolds Trimax Pen - BLUE, BLACK, RED & GREEN Gel Pen		29
Reynolds Brite BP Pen Ball Pen		49
Reynolds JIFFY Blue Gel Pen		3
Reynolds Brite Blue Pen Packet Ball Pen		0
Reynolds D Fine Ball Pen		12
Reynolds Etek Roller Ball Pen		32

Item	#	Units
Reynolds Feltex Roller Ball Pen		32
Reynolds Trimax Pens Pouch - Blue, Black, Red & Green Gel Pen		29
Reynolds SmothMate Ball Pen		14
Reynolds 045 Pen Jar Ball Pen		4
Reynolds JIFFY Gel Pen		11
FLAIR Silkin Jar of Ball Pen		6
Cello Butterflow Simply Ball Pen		0
Classmate Pulse 6 Subject (240 X 180 mm) Spiral B5 Notebook Single Line 300 Pages		12
Classmate Pulse 1 Subject (240 X 180 mm) Spiral B5 Notebook Single Line 180 Pages		0
NAVNEET Youva Soft Bound Brown Note Book 18x24 cm Single Line 92 Pages Regular Note Book		28
Classmate Pulse 1 Subject (267 X 203 mm) Spiral Regular Notebook Single Line 160 Pages		39
Classmate Longbook A4 Notebook Unruled 140 Pages		13
Classmate Longbook A4 Notebook Single Line 240 Pages		0
CASIO FX-991MS Scientific Calculator		4

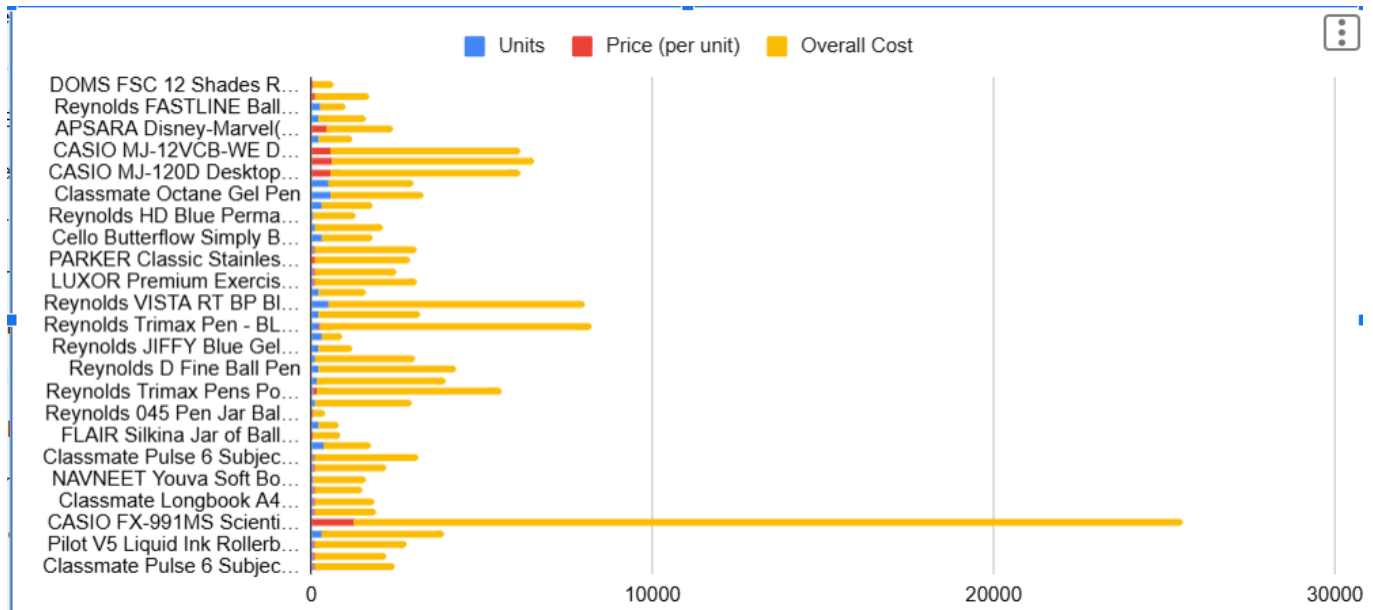
DESCRIPTIVE STATISTICS:-

After processing and cleansing the data, the following is a summarized overview of the dataset using descriptive statistics. I have focused on key insights, highlighting the most relevant descriptive statistics for both sales and inventory data.

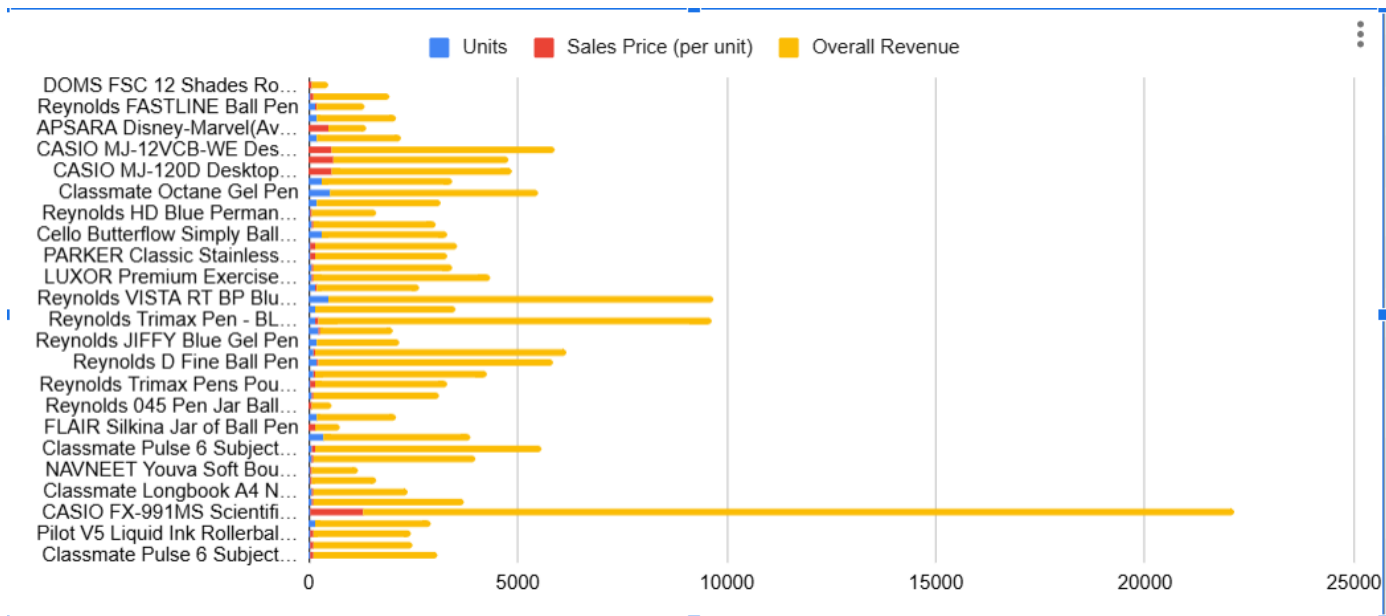
Measure	Definition
Sum	Total items sold
Mean	Average items sold
Median	Middle value
Minimum	Least revenue item
Maximum	Maximum revenue item
Standard Deviation	The measure of spread of values from mean in single sample.

Measure	Quantity	Total Revenue
Sum	5244	160279
Mean	116.533	3561.755
Median	68	3000
Minimum	2	406
Maximum	500	20800
Standard Deviation	120.883	3250.255

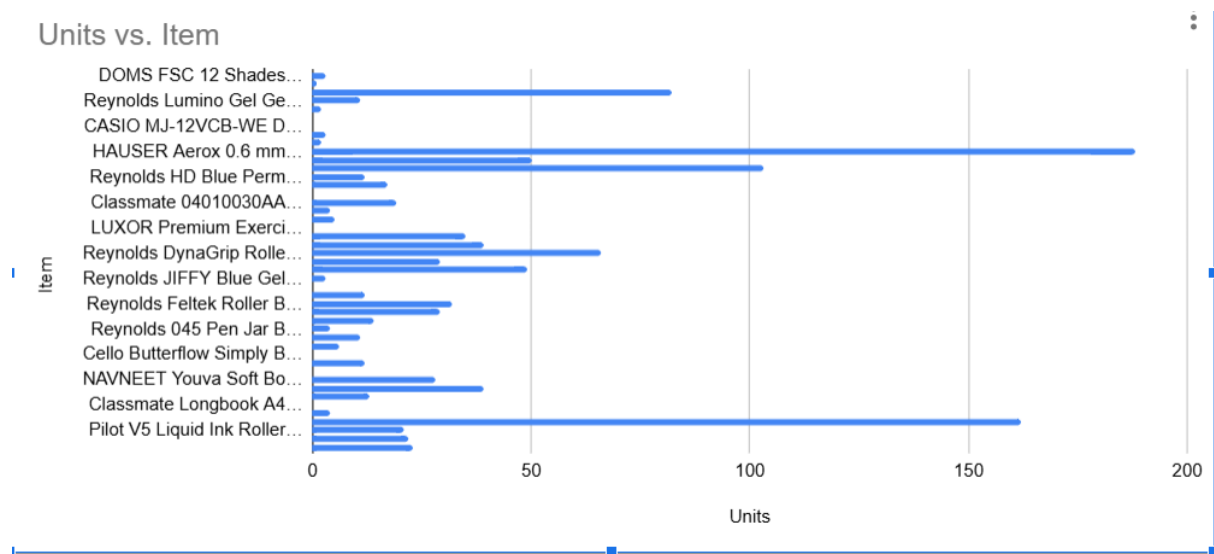
Purchase



Sales



Inventory



Detailed Explanation of Analysis Process & Methods:-

1. Data Collection & Understanding the Business:

The analysis began with collecting past transaction records, sales data, and inventory details from Aren's Fancy World, a stationery and gift shop. A short interview with the owner provided insights into key challenges, such as inventory management, customer retention, and handling special order requests that often lead to surplus stock.

2. Quantitative Analysis Methods:

- **Spreadsheets & Excel Tools:** Used for calculating total credit across customers, analysing sales trends, and computing costs. Functions like **SUM**, **AVERAGE**, **Median**, **Minimum**, **Maximum**, **Standard Deviation** and **Charts**.

3. Qualitative Insights from Business Owner:

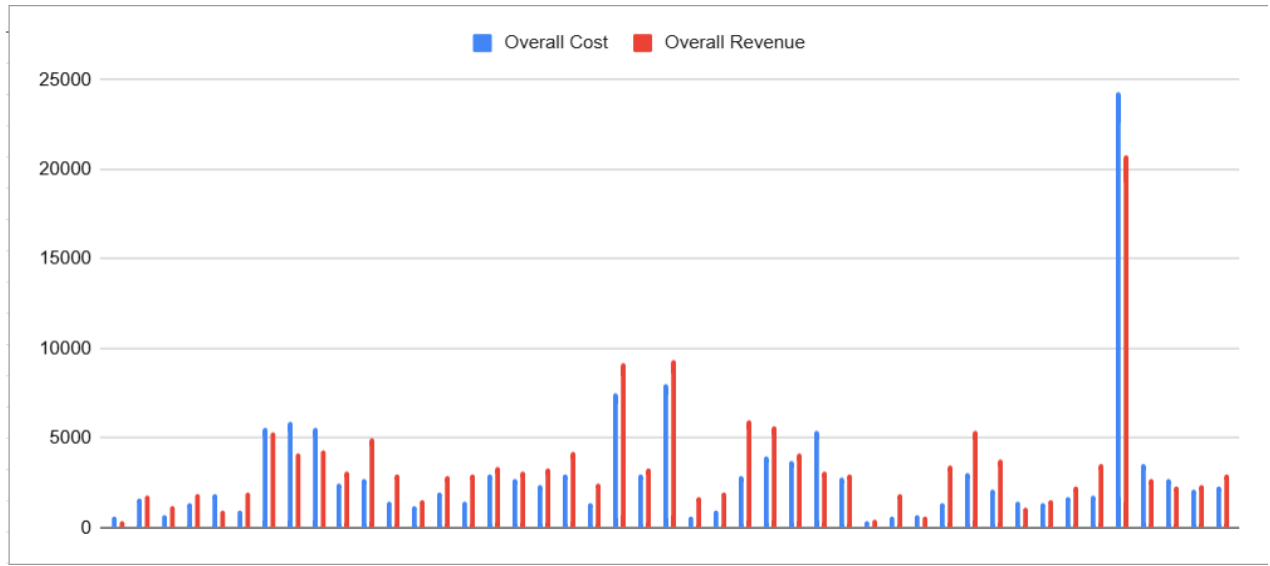
Conversations with the owner revealed additional concerns, including customer returns, difficulty in selling bulk-ordered items, and competition from online platforms. Possible solutions included introducing new product categories, exploring online sales channels, and implementing stricter return policies.

4. Key Findings & Recommendations:

- Improve **inventory forecasting** using past sales trends.
- Introduce **return policies** to minimize loss from customer returns.
- Explore **online selling platforms** for better market reach.
- Implement **bulk order tracking** to ensure demand before stocking special items.

Results and Findings:-

The key observations on the net sales and the net revenue:



A consistent pattern can be seen between the sales and the cost of items.

For most of the items the revenue generated is more than the cost of the item.

Maximum profit is earned on gel pens and ball pens, and the least revenue is generated on packets or jars of pens.

The large value items should be carefully managed as they contribute most to the revenue but also significantly impact the overall cost.

- Total Profit = Rs. 23,159
- Total Revenue = Rs. 1,60,279
- Mean Revenue = Rs. 3,561.755
- Median Revenue = Rs. 3,000
- Standard Deviation = Rs. 3,250.255